

3.2.8 Specialist techniques and processes

In relation to **at least one** material category or system, students should know and understand the factors listed below.

The use of production aids

Content	Potential links to maths and science
How to use measurement/reference points, templates, jigs and patterns where suitable.	Scaling of drawings, working to datums. Material quantities required.

Tools, equipment and processes

Content	Potential links to maths and science
A range of tools, equipment and processes that can be used to shape, fabricate, construct and assemble high quality prototypes, as appropriate to the materials and/or components being used including: wastage, such as: • die cutting • perforation • turning • sawing • milling • drilling • cutting and shearing addition, such as: • brazing • welding • lamination • soldering • 3D printing • batik • sewing • bonding • printing deforming and reforming such as: • vacuum forming • creasing • pressing • drape forming • bending • folding • blow moulding • casting • injection moulding • extrusion.	

How materials are cut shaped and formed to a tolerance

Content	Potential links to maths and science
The manufacture to minimum and maximum measurements.	Extracting information on tolerances and using it to control quality and make a prototype.

Commercial processes

Content	Potential links to maths and science
• Papers and boards (offset lithography and die cutting). • Timber based materials (routing and turning). • Metal based materials (milling and casting). • Polymers (injection molding and extrusion). • Textile based materials (weaving, dying and printing). • Electrical and mechanical systems (pick and place assembly and flow soldering).	

Quality control

Content	Potential links to maths and science
The application and use of quality control to include measurable and quantitative systems used during manufacture • Papers and boards (registration marks). • Timber based materials (dimensional accuracy using go/no go fixture). • Metal based materials (dimensional accuracy using a depth stop). • Polymers (dimensional accuracy by selecting correct laser settings). • Textile based materials (dimensional accuracy checking a repeating print against an original sample). • Electrical and mechanical systems (UV exposure, developing and etching times in PCB manufacture).	